Lesson Planning

Complete Lesson Plan Book

Class: Class 12 – ICS

Subject: Computer Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Software Concepts

Learning Objectives

• Define and explain system software in the context of Software Concepts.

• Differentiate between application software and software development life cycle with suitable examples.

• Solve conceptual problems related to software quality accurately.

• Apply the ideas of Software Concepts to everyday situations involving open source software.

Key Concepts

• Meaning, definition and scope of system software.

• Important properties, types and examples of application software.

• Relationship between software development life cycle and software quality.

• Practical applications of Software Concepts, especially open source software.

• Key terms and a summary of the main points of Software Concepts.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Software Concepts with a daily-life example.
2. Explain system software on the board with definitions and clear diagrams.
3. Discuss application software and software development life cycle using simple, relatable examples.
4. Involve the students through short questions on software quality.
5. Solve an exercise question on open source software together with the class.
6. Summarize the lesson and highlight the key points of Software Concepts.

Classroom Activities

• Group discussion: students compare application software with software development life cycle.

• Board activity: solve a question on software quality in pairs.

• Quick quiz on the key terms of Software Concepts.

Assessment Questions

1. Define system software.
2. Differentiate between application software and software development life cycle.
3. Explain with an example: software quality.
4. Write a short note on open source software.

Homework

• Read Software Concepts from the book and solve the exercise questions.

• Prepare short notes on system software and application software.

Time Allocation

4–6 periods of 40 minutes each

Unit 2: Data Basics

Learning Objectives

• Define and explain data vs information in the context of Data Basics.
• Differentiate between data types and data hierarchy with suitable examples.
• Solve conceptual problems related to data integrity accurately.
• Apply the ideas of Data Basics to everyday situations involving data security.

Key Concepts

• Meaning, definition and scope of data vs information.
• Important properties, types and examples of data types.
• Relationship between data hierarchy and data integrity.
• Practical applications of Data Basics, especially data security.
• Key terms and a summary of the main points of Data Basics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Data Basics with a daily-life example.
2. Explain data vs information on the board with definitions and clear diagrams.
3. Discuss data types and data hierarchy using simple, relatable examples.
4. Involve the students through short questions on data integrity.
5. Solve an exercise question on data security together with the class.
6. Summarize the lesson and highlight the key points of Data Basics.

Classroom Activities

• Group discussion: students compare data types with data hierarchy.
• Board activity: solve a question on data integrity in pairs.
• Quick quiz on the key terms of Data Basics.

Assessment Questions

1. Define data vs information.
2. Differentiate between data types and data hierarchy.
3. Explain with an example: data integrity.
4. Write a short note on data security.

Homework

• Read Data Basics from the book and solve the exercise questions.
• Prepare short notes on data vs information and data types.

Time Allocation

4–6 periods of 40 minutes each

Unit 3: System Analysis and Design

Learning Objectives

• Define and explain system analyst role in the context of System Analysis and Design.
• Differentiate between feasibility study and data flow diagrams with suitable examples.
• Solve conceptual problems related to system design accurately.
• Apply the ideas of System Analysis and Design to everyday situations involving system testing.

Key Concepts

• Meaning, definition and scope of system analyst role.
• Important properties, types and examples of feasibility study.
• Relationship between data flow diagrams and system design.
• Practical applications of System Analysis and Design, especially system testing.
• Key terms and a summary of the main points of System Analysis and Design.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce System Analysis and Design with a daily-life example.
2. Explain system analyst role on the board with definitions and clear diagrams.
3. Discuss feasibility study and data flow diagrams using simple, relatable examples.
4. Involve the students through short questions on system design.
5. Solve an exercise question on system testing together with the class.
6. Summarize the lesson and highlight the key points of System Analysis and Design.

Classroom Activities

• Group discussion: students compare feasibility study with data flow diagrams.
• Board activity: solve a question on system design in pairs.
• Quick quiz on the key terms of System Analysis and Design.

Assessment Questions

1. Define system analyst role.
2. Differentiate between feasibility study and data flow diagrams.
3. Explain with an example: system design.
4. Write a short note on system testing.

Homework

• Read System Analysis and Design from the book and solve the exercise questions.
• Prepare short notes on system analyst role and feasibility study.

Time Allocation

4–6 periods of 40 minutes each

Unit 4: Database Fundamentals

Learning Objectives

• Define and explain database concepts in the context of Database Fundamentals.
• Differentiate between DBMS advantages and data models with suitable examples.
• Solve conceptual problems related to keys accurately.
• Apply the ideas of Database Fundamentals to everyday situations involving normalization.

Key Concepts

• Meaning, definition and scope of database concepts.
• Important properties, types and examples of DBMS advantages.
• Relationship between data models and keys.
• Practical applications of Database Fundamentals, especially normalization.
• Key terms and a summary of the main points of Database Fundamentals.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Database Fundamentals with a daily-life example.
2. Explain database concepts on the board with definitions and clear diagrams.
3. Discuss DBMS advantages and data models using simple, relatable examples.
4. Involve the students through short questions on keys.
5. Solve an exercise question on normalization together with the class.
6. Summarize the lesson and highlight the key points of Database Fundamentals.

Classroom Activities

• Group discussion: students compare DBMS advantages with data models.
• Board activity: solve a question on keys in pairs.
• Quick quiz on the key terms of Database Fundamentals.

Assessment Questions

1. Define database concepts.
2. Differentiate between DBMS advantages and data models.
3. Explain with an example: keys.
4. Write a short note on normalization.

Homework

• Read Database Fundamentals from the book and solve the exercise questions.
• Prepare short notes on database concepts and DBMS advantages.

Time Allocation

4–6 periods of 40 minutes each

Unit 5: Getting Started with DBMS

Learning Objectives

• Define and explain DBMS interface in the context of Getting Started with DBMS.

• Differentiate between creating tables and data types in tables with suitable examples.

• Solve conceptual problems related to primary key accurately.

• Apply the ideas of Getting Started with DBMS to everyday situations involving saving databases.

Key Concepts

• Meaning, definition and scope of DBMS interface.

• Important properties, types and examples of creating tables.

• Relationship between data types in tables and primary key.

• Practical applications of Getting Started with DBMS, especially saving databases.

• Key terms and a summary of the main points of Getting Started with DBMS.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Getting Started with DBMS with a daily-life example.
2. Explain DBMS interface on the board with definitions and clear diagrams.
3. Discuss creating tables and data types in tables using simple, relatable examples.
4. Involve the students through short questions on primary key.
5. Solve an exercise question on saving databases together with the class.
6. Summarize the lesson and highlight the key points of Getting Started with DBMS.

Classroom Activities

• Group discussion: students compare creating tables with data types in tables.

• Board activity: solve a question on primary key in pairs.

• Quick quiz on the key terms of Getting Started with DBMS.

Assessment Questions

1. Define DBMS interface.
2. Differentiate between creating tables and data types in tables.
3. Explain with an example: primary key.
4. Write a short note on saving databases.

Homework

• Read Getting Started with DBMS from the book and solve the exercise questions.

• Prepare short notes on DBMS interface and creating tables.

Time Allocation

4–6 periods of 40 minutes each

Unit 6: Tables and Queries

Learning Objectives

• Define and explain table relationships in the context of Tables and Queries.
• Differentiate between query design and select queries with suitable examples.
• Solve conceptual problems related to criteria accurately.
• Apply the ideas of Tables and Queries to everyday situations involving sorting and filtering.

Key Concepts

• Meaning, definition and scope of table relationships.
• Important properties, types and examples of query design.
• Relationship between select queries and criteria.
• Practical applications of Tables and Queries, especially sorting and filtering.
• Key terms and a summary of the main points of Tables and Queries.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Tables and Queries with a daily-life example.
2. Explain table relationships on the board with definitions and clear diagrams.
3. Discuss query design and select queries using simple, relatable examples.
4. Involve the students through short questions on criteria.
5. Solve an exercise question on sorting and filtering together with the class.
6. Summarize the lesson and highlight the key points of Tables and Queries.

Classroom Activities

• Group discussion: students compare query design with select queries.
• Board activity: solve a question on criteria in pairs.
• Quick quiz on the key terms of Tables and Queries.

Assessment Questions

1. Define table relationships.
2. Differentiate between query design and select queries.
3. Explain with an example: criteria.
4. Write a short note on sorting and filtering.

Homework

• Read Tables and Queries from the book and solve the exercise questions.
• Prepare short notes on table relationships and query design.

Time Allocation

4–6 periods of 40 minutes each

Unit 7: Forms and Reports

Learning Objectives

• Define and explain form design in the context of Forms and Reports.
• Differentiate between form controls and report design with suitable examples.
• Solve conceptual problems related to report sections accurately.
• Apply the ideas of Forms and Reports to everyday situations involving printing reports.

Key Concepts

• Meaning, definition and scope of form design.
• Important properties, types and examples of form controls.
• Relationship between report design and report sections.
• Practical applications of Forms and Reports, especially printing reports.
• Key terms and a summary of the main points of Forms and Reports.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Forms and Reports with a daily-life example.
2. Explain form design on the board with definitions and clear diagrams.
3. Discuss form controls and report design using simple, relatable examples.
4. Involve the students through short questions on report sections.
5. Solve an exercise question on printing reports together with the class.
6. Summarize the lesson and highlight the key points of Forms and Reports.

Classroom Activities

• Group discussion: students compare form controls with report design.
• Board activity: solve a question on report sections in pairs.
• Quick quiz on the key terms of Forms and Reports.

Assessment Questions

1. Define form design.
2. Differentiate between form controls and report design.
3. Explain with an example: report sections.
4. Write a short note on printing reports.

Homework

• Read Forms and Reports from the book and solve the exercise questions.
• Prepare short notes on form design and form controls.

Time Allocation

4–6 periods of 40 minutes each

Unit 8: C Language: Elements and Program Structure

Learning Objectives

• Define and explain C tokens in the context of C Language: Elements and Program Structure.

• Differentiate between constants and escape sequences with suitable examples.

• Solve conceptual problems related to program layout accurately.

• Apply the ideas of C Language: Elements and Program Structure to everyday situations involving debugging.

Key Concepts

• Meaning, definition and scope of C tokens.

• Important properties, types and examples of constants.

• Relationship between escape sequences and program layout.

• Practical applications of C Language: Elements and Program Structure, especially debugging.

• Key terms and a summary of the main points of C Language: Elements and Program Structure.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce C Language: Elements and Program Structure with a daily-life example.

2. Explain C tokens on the board with definitions and clear diagrams.

3. Discuss constants and escape sequences using simple, relatable examples.

4. Involve the students through short questions on program layout.

5. Solve an exercise question on debugging together with the class.

6. Summarize the lesson and highlight the key points of C Language: Elements and Program Structure.

Classroom Activities

• Group discussion: students compare constants with escape sequences.

• Board activity: solve a question on program layout in pairs.

• Quick quiz on the key terms of C Language: Elements and Program Structure.

Assessment Questions

1. Define C tokens.

2. Differentiate between constants and escape sequences.

3. Explain with an example: program layout.

4. Write a short note on debugging.

Homework

• Read C Language: Elements and Program Structure from the book and solve the exercise questions.

• Prepare short notes on C tokens and constants.

Time Allocation

4–6 periods of 40 minutes each

Unit 9: Functions and Subprograms in C

Learning Objectives

• Define and explain function prototypes in the context of Functions and Subprograms in C.
• Differentiate between parameters and return values with suitable examples.
• Solve conceptual problems related to local variables accurately.
• Apply the ideas of Functions and Subprograms in C to everyday situations involving recursion.

Key Concepts

• Meaning, definition and scope of function prototypes.
• Important properties, types and examples of parameters.
• Relationship between return values and local variables.
• Practical applications of Functions and Subprograms in C, especially recursion.
• Key terms and a summary of the main points of Functions and Subprograms in C.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Functions and Subprograms in C with a daily-life example.
2. Explain function prototypes on the board with definitions and clear diagrams.
3. Discuss parameters and return values using simple, relatable examples.
4. Involve the students through short questions on local variables.
5. Solve an exercise question on recursion together with the class.
6. Summarize the lesson and highlight the key points of Functions and Subprograms in C.

Classroom Activities

• Group discussion: students compare parameters with return values.
• Board activity: solve a question on local variables in pairs.
• Quick quiz on the key terms of Functions and Subprograms in C.

Assessment Questions

1. Define function prototypes.
2. Differentiate between parameters and return values.
3. Explain with an example: local variables.
4. Write a short note on recursion.

Homework

• Read Functions and Subprograms in C from the book and solve the exercise questions.
• Prepare short notes on function prototypes and parameters.

Time Allocation

4–6 periods of 40 minutes each

Unit 10: File Handling in C

Learning Objectives

• Define and explain file concept in the context of File Handling in C.
• Differentiate between file modes and reading files with suitable examples.
• Solve conceptual problems related to writing files accurately.
• Apply the ideas of File Handling in C to everyday situations involving file functions.

Key Concepts

• Meaning, definition and scope of file concept.
• Important properties, types and examples of file modes.
• Relationship between reading files and writing files.
• Practical applications of File Handling in C, especially file functions.
• Key terms and a summary of the main points of File Handling in C.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce File Handling in C with a daily-life example.
2. Explain file concept on the board with definitions and clear diagrams.
3. Discuss file modes and reading files using simple, relatable examples.
4. Involve the students through short questions on writing files.
5. Solve an exercise question on file functions together with the class.
6. Summarize the lesson and highlight the key points of File Handling in C.

Classroom Activities

• Group discussion: students compare file modes with reading files.
• Board activity: solve a question on writing files in pairs.
• Quick quiz on the key terms of File Handling in C.

Assessment Questions

1. Define file concept.
2. Differentiate between file modes and reading files.
3. Explain with an example: writing files.
4. Write a short note on file functions.

Homework

• Read File Handling in C from the book and solve the exercise questions.
• Prepare short notes on file concept and file modes.

Time Allocation

4–6 periods of 40 minutes each